Maximum-Length Sequence for Impulse Testing of the Ionosphere (WWV WG)

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Approach

- Incorporate a 5-10 second audio file of a repeated Maximum Length Sequence (MLS) into a WWV experimental waveform.
- Matlab script generates a repeated 511-bit MLS at a base data rate of 6kbs.
- The MLS is filtered and upsampled using a root-raised cosine filter (RRC).
- The filtered waveform is stored off as a 48 kHz .wav file for incorporation into a future WWV experimental waveform
- Resulting waveform has smooth transitions and a BW of 3.1 kHz

Reception of Over-the-Air WWV Transmissions

- SDR or Ham transceiver set for USB reception (BW>2700 Hz) of WWV signal
- Frequency reference (e.g. Bodnar GPSDO) recommended
- Record the received audio file using Audacity or other audio recording software and save as a .wav file
- Use post processing described in my NRSM paper and HamSCI 2026 presentation (see last slide)
- The post processing software is currently a prototype Matlab script that needs cleaning up for general distribution. Scripts could be converted to Python as well for dissemination

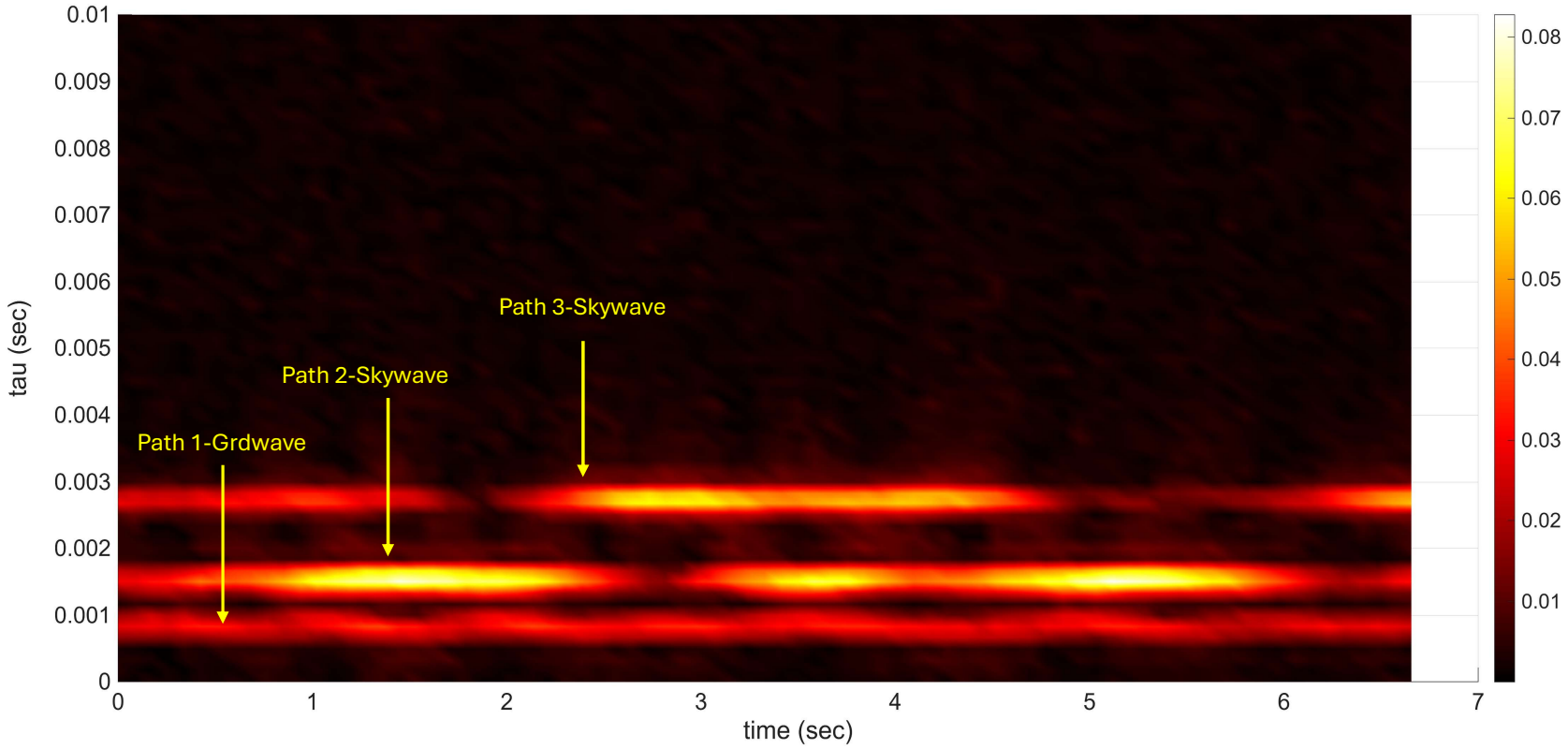
Proposed Early Experiments

- Generate protype waveform within the WWV WG and submit for incorporation into the updated WWV experimental waveform
- Would like to perform proof-of-concept tests over a trial period to test out the efficacy of the approach
- This technique has been extensively tested for single-sideband transmissions
- Need to test with AM modulation transmissions from WWV
- Used ham transceivers and remote SDRs to collect early test results

Baseband Channel Simulation Results + Plots

- Repeated 511-bit MLS (80 repetitions), 6.8 sec duration,
- 3-path Watterson channel model—Baseband simulation
- Path 1: Ground wave, Doppler spread=0.02 Hz
- Path 2: Skywave, 0.7ms delay, Doppler spread=0.02 Hz, Doppler offset = -1.0 Hz
- Path 3: Skywave, 1.9ms delay, Doppler spread=0.02 Hz, Doppler offset = 0.5 Hz
- Note: Time delays are relative to Path 1
- Plots derived from measured channel matrix: delay-time, delay-Doppler, time-channel frequency response
- Bello functions expressing time, delay, Doppler, and channel frequency response
- This is a relative time measurement technique.

Time Varying Channel Impulse Response

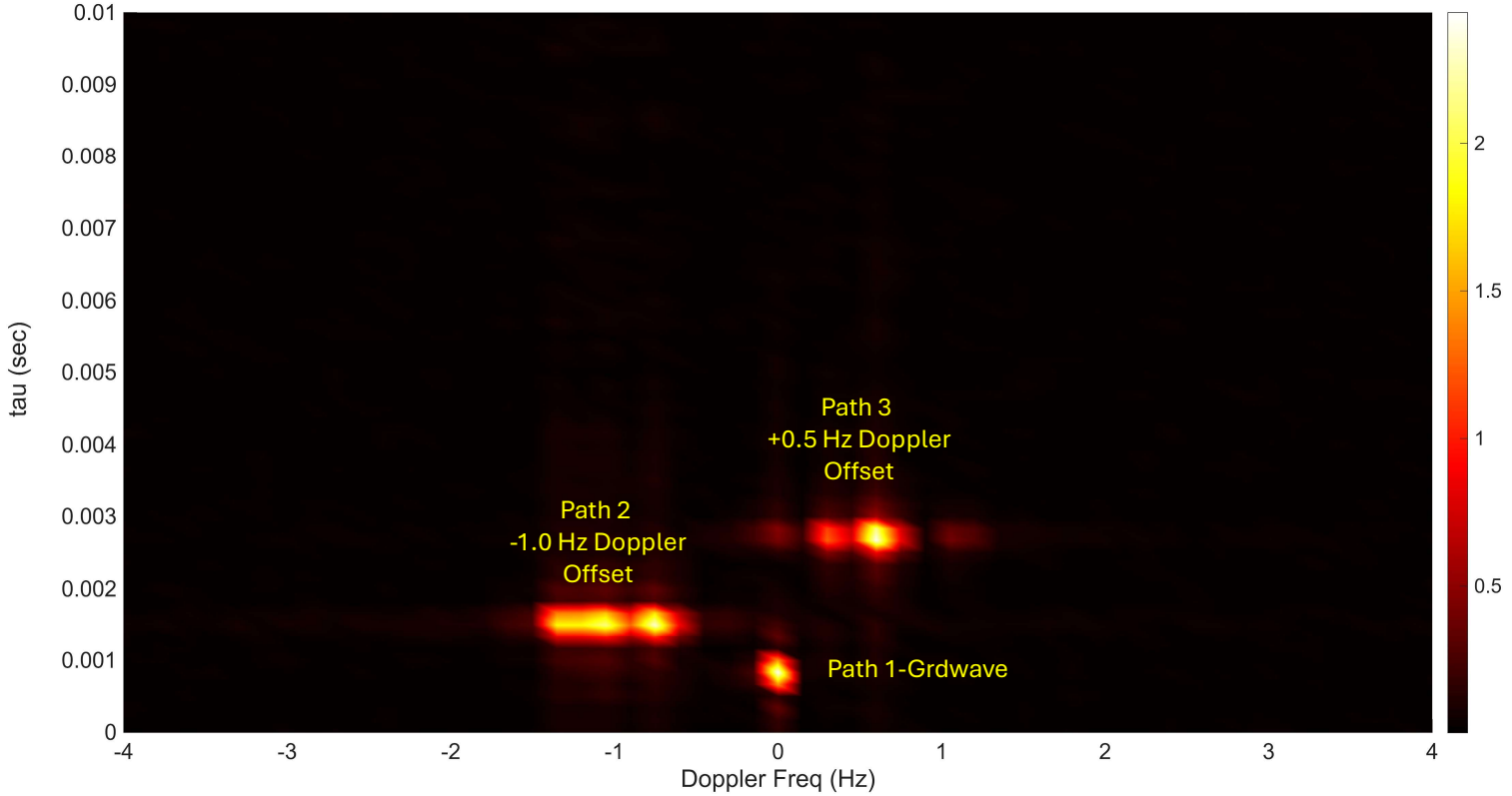


Time-Variability Video Channel Impulse Response

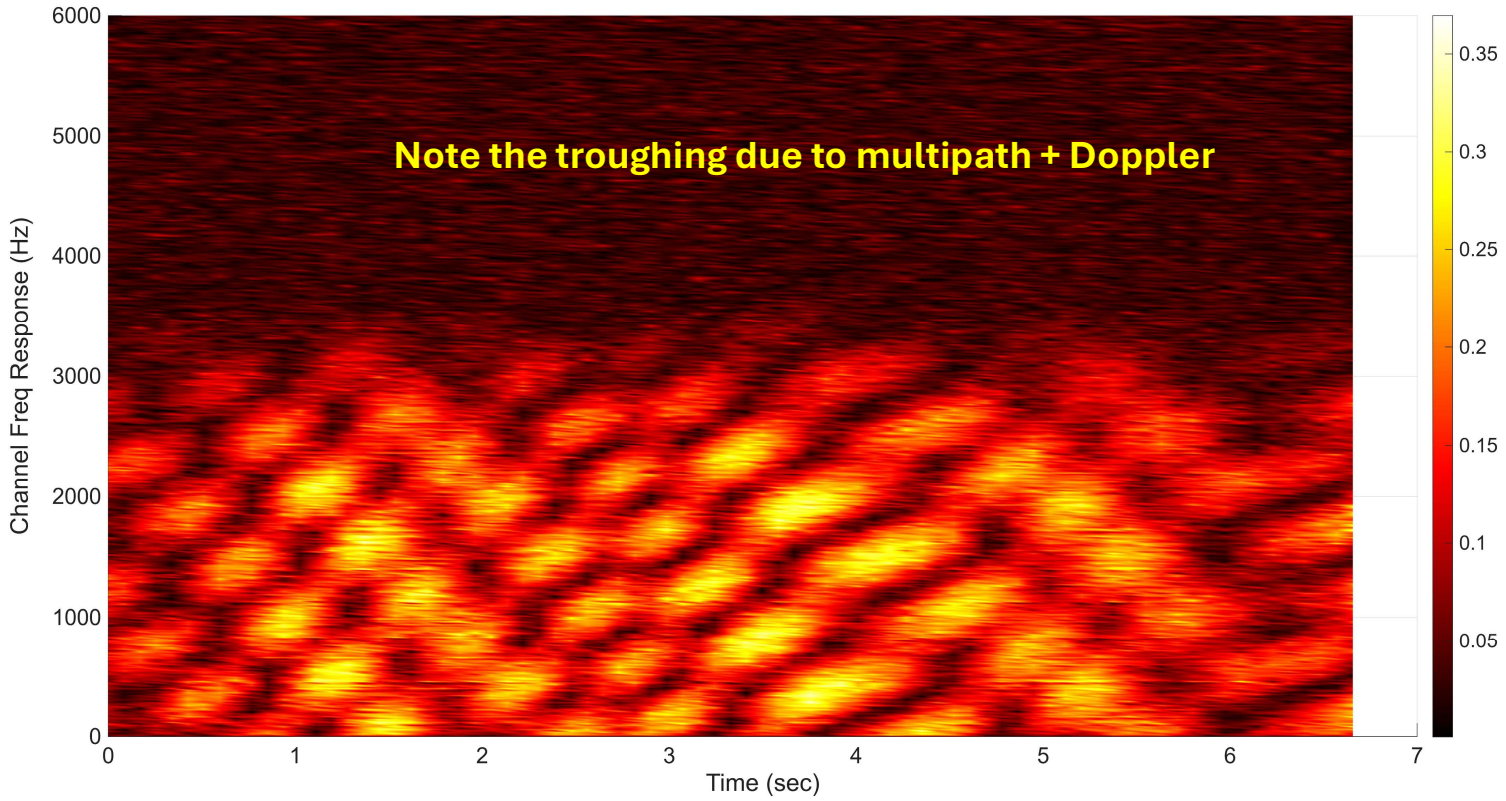
- Observed time variability for a 6.8 s MLS sequence
- Video link: https://youtu.be/SwHV4r-hjBM
- This is an animation obtained directly form the measured channel matrix
- This simulates what a post-processed capture from a WWV OTA waveform might look like
- Note the time-of-flight information obtained from a 6.8 s capture
- Path 2 response occurs approximately 0.7 ms after Path 1 & Path 3 response 1.9 ms after Path 1 arrival

Delay Doppler using a 6.8 s record

(Note: Doppler resolution improves with longer records)



Channel Frequency Response



References

- Summary paper for NRSM presentation in January 2026: https://usncursi.org/archive/nrsm/2026/papers/1011.pdf
- My HamSCI 2026 presentation: https://www.youtube.com/watch?v=p5obYvQ-hlQ
- HamSCI Publication Link: https://hamsci.org/publications/amateur-radio-hf-channel-sounder